FORWARD DEPLOYED ENGINEER

SMART Curriculum V3.0 | 12 weeks | 15 hours/week | Mastery-gated

Focus: AI x Risk/Fraud x Enterprise Integrations. One evolving Enterprise Fraud Investigation AI Copilot, built with synthetic financial data.

01 / WEEKLY CURRICULUM

Week	Competency	Study scope	Resource(s)	Proof of mastery
1	Python production	Typing, dataclasses/Pydantic, exceptions, pytest, async basics; skip beginner syntax	Python tutorial; pytest	Typed fraud-rule engine; unseen tests + error-handling defense
2	APIs and webhooks	FastAPI routes, schemas, authentication dependencies, signatures, idempotency, retries	FastAPI tutorial; Stripe Webhooks	Merchant event ingestion; invalid signatures and duplicates rejected
3	SQL and data	Joins, CTEs, windows, indexes, transactions, EXPLAIN, basic ETL	PostgreSQL tutorial; SQLBolt	Investigation database + 10 verified analytical queries
4	Security and tenancy	OAuth2/JWT basics, RBAC, tenant isolation, secrets, OWASP API risks	OWASP API Security; FastAPI Security	Unauthorized access and cross-tenant isolation tests pass
5	LLM engineering	Structured outputs, tool calls, prompts, context, errors, token costs	OpenAI API docs; OpenAI Cookbook	Case summarizer with evidence references and validated output
6	RAG and retrieval	Chunking, embeddings, retrieval, filters, retrieval metrics; skip fine-tuning	LangChain RAG tutorial	Policy retrieval with labeled recall and grounded-answer evaluation
7	Agent workflows	Tool orchestration, state, checkpoints, bounded execution, approval, recovery	Hugging Face Agents; LangGraph	Auditable fraud investigation workflow with controlled tools
8	AI evaluations	Golden sets, regressions, injection attacks, permissions, human review	LangSmith; OWASP GenAI	50-case suite; zero critical unauthorized actions
9	Deployment	Docker, Compose, CI/CD, cloud deploy, config and secrets; skip Kubernetes	Docker Get Started; AWS Skill Builder	Reproducible authenticated cloud demo and CI checks
10	Reliability	Timeouts, retries, queues, DLQ, logs, metrics, tracing, incidents	Google SRE Book; OpenTelemetry	Diagnose injected faults and demonstrate recovery
11	Customer solutioning	Discovery, stakeholder map, scoping, architecture, ROI assumptions, writing	AWS Well-Architected; simulated cases	Requirements, proposal, architecture and 10-minute presentation
12	Capstone validation	Full integration, customer demo, production simulation, documentation	GitHub; external review; mock interviews	Working solution, tests, live demo and design defense

02 / WEEKLY EXECUTION RHYTHM

Day	Hours	Activity
Monday	2h	Bounded concepts and examples
Tuesday	2h	Independent implementation
Wednesday	2h	Exercises and tests

Day	Hours	Activity
Thursday	2h	Capstone extension and debugging
Friday	2h	Unseen challenge and oral defense
Saturday	3h	Customer simulation and integration
Sunday	2h	Mastery exam and spaced revision

Total: 15 hours/week, ~180 hours over 12 weeks. Mastery gates override calendar dates.

03 / ASSESSMENT SYSTEM

Assessment	Timing	Pass criteria
Concept quiz	After each module	80% or higher on unseen questions
Coding	Weekly	Critical tests pass; no serious security flaws
Debugging	Weekly from week 2	Root cause, fix and regression test
Architecture defense	Weeks 4, 8, 12	Explain trade-offs and failure modes
Retention	7 and 21 days later	80% or higher on equivalent unseen questions
Customer simulation	Weeks 4, 8, 11, 12	Clarify scope, explain trade-offs and outcomes
Independent review	Week 12	Reviewer reproduces, tests and evaluates system

Five gates: Knowledge > Application > Transfer > Defense > Retention. Retest failed critical gates.

04 / CAPSTONE

Merchant integration (APIs/webhooks) > Fraud data (SQL/RBAC) > AI copilot (LLM/RAG/tools) > Operations (Docker/AWS/tracing) > Customer delivery (proposal/demo).

05 / EXCLUSIONS & QUALITY CONTROLS

Skip Kubernetes administration, model training, advanced ML mathematics, multiple agent frameworks, exhaustive AWS services, and unrelated frontend complexity. Study only bounded concepts, not entire documentation sites.

Use automated tests, unseen scenarios, reproducible artifacts and an independent reviewer. AI feedback alone is not proof of mastery. Never use confidential employer or customer data.

This roadmap is execution-ready as a plan; exact lesson subsections, hidden fixtures, answer keys and detailed grading rubrics still need to be prepared for a fully self-contained course.

RESOURCE LINKS

Python tutorial — https://docs.python.org/3/tutorial/

pytest — https://docs.pytest.org/en/stable/getting-started.html

FastAPI tutorial — https://fastapi.tiangolo.com/tutorial/

Stripe Webhooks — https://docs.stripe.com/webhooks

PostgreSQL tutorial — https://www.postgresql.org/docs/current/tutorial.html

SQLBolt — https://sqlbolt.com/

OWASP API Security — https://owasp.org/API-Security/

FastAPI Security — https://fastapi.tiangolo.com/tutorial/security/

OpenAI API docs — https://platform.openai.com/docs

OpenAI Cookbook — https://cookbook.openai.com/

LangChain RAG tutorial — https://docs.langchain.com/oss/python/learn

Hugging Face Agents — https://huggingface.co/learn/agents-course/

LangGraph — https://docs.langchain.com/oss/python/langgraph/overview

LangSmith — https://docs.langchain.com/langsmith/

OWASP GenAI — https://genai.owasp.org/

Docker Get Started — https://docs.docker.com/get-started/

AWS Skill Builder — https://skillbuilder.aws/

Google SRE Book — https://sre.google/sre-book/table-of-contents/

OpenTelemetry — https://opentelemetry.io/docs/

AWS Well-Architected — https://aws.amazon.com/architecture/well-architected/

GitHub — https://github.com/